# Reward Modeling for Zen Alignment: Multi-Reward Ensembles and Synthetic Preference Generation

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**Abstract**

Reward modeling is a critical component of aligning large language models to human preferences. We present the Zen alignment stack, comprising: (1) a preference data collection pipeline that produces high-quality human judgments at scale; (2) ensemble reward models based on the Bradley-Terry statistical model with multi-head architecture; (3) synthetic preference generation using a red-team/blue-team approach to augment human data; and (4) Direct Preference Optimization (DPO) as an alternative to full RL training. Our multi-reward ensemble achieves 87.3% correlation with human judgment on a held-out evaluation set, a 12% improvement over single-model baselines. RLHF with our ensemble reward substantially outperforms DPO alone on open-ended generation quality, while DPO achieves competitive results at lower compute cost for instruction following.

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# 1 Introduction

Training a large language model to generate text that humans find helpful, harmless, and honest (HHH) [1] requires a reward signal that accurately captures human preferences. Reinforcement Learning from Human Feedback (RLHF) [2, 3] trains a reward model on preference data and then uses it to fine-tune the policy via PPO. The quality of the reward model is a critical bottleneck: a poorly calibrated reward model produces *reward hacking*, where the policy exploits reward model weaknesses rather than genuinely improving.

We address reward model quality through three mechanisms:

1. **High-quality preference data**: structured collection of pairwise human judgments with inter-annotator agreement metrics.
2. **Multi-reward ensemble**: an ensemble of reward models with different training objectives (helpfulness, harmlessness, honesty, instruction-following), combined to produce a holistic alignment score.
3. **Synthetic preference generation**: using a red-team model to generate challenging response pairs that expose alignment failures, augmenting human data with synthetically generated hard cases.

## 1.1 Alignment Taxonomy

Following [1], we distinguish:

- **Helpfulness**: whether the response addresses the user's intent effectively.
- **Harmlessness**: whether the response avoids harmful content or enabling harm.
- **Honesty**: whether the response is factually accurate and appropriately uncertain.
- **Instruction following**: whether the response adheres to specified format, length, and style constraints.

These dimensions are partially correlated but not identical, motivating a multi-head reward architecture.

# 2 Preference Data Collection

## 2.1 Annotation Protocol

Human annotators compare pairs of responses $(y_A, y_B)$ to the same prompt $x$ and indicate: strongly prefer $A$, slightly prefer $A$, about equal, slightly prefer $B$, strongly prefer $B$. This 5-point scale is mapped to a continuous preference signal $\delta \in \{-2, -1, 0, +1, +2\}$ for the Bradley-Terry likelihood computation.

Annotators also rate each response on four dimensions (helpfulness, harmlessness, honesty, instruction-following) using a 1–5 Likert scale. These dimensional ratings train the individual reward model heads.

## 2.2 Inter-Annotator Agreement

To ensure data quality, each prompt-pair is annotated by 3 independent annotators. We compute Fleiss's $\kappa$ as a quality metric:

Table 1: Inter-annotator agreement across preference data categories.

| Category | Fleiss's $\kappa$ | Agreement interpretation |
|---|---|---|
| Helpfulness | 0.68 | Substantial |
| Harmlessness | 0.81 | Almost perfect |
| Honesty | 0.71 | Substantial |
| Instruction following | 0.87 | Almost perfect |
| Overall preference | 0.64 | Substantial |

Prompt-pairs with $\kappa < 0.4$ are excluded from training, retaining 89% of collected pairs. This yields a final dataset of 1.2M preference pairs.

### 2.3 Dataset Composition

Table 2: Preference dataset composition by domain.

| Domain | Pairs (K) | Fraction (%) | Avg. $\kappa$ |
|---|---|---|---|
| General conversation | 320 | 26.7 | 0.63 |
| Instruction following | 210 | 17.5 | 0.88 |
| Mathematics / reasoning | 180 | 15.0 | 0.82 |
| Code generation | 150 | 12.5 | 0.79 |
| Factual QA | 140 | 11.7 | 0.71 |
| Safety-critical | 120 | 10.0 | 0.84 |
| Creative writing | 80 | 6.7 | 0.58 |
| **Total** | 1200 | 100.0 | 0.74 |

## 3 Bradley-Terry Reward Model

### 3.1 Bradley-Terry Model

The Bradley-Terry (BT) model [4] models the probability that response $y_A$ is preferred over $y_B$ for prompt $x$ as:

$$P(y_A \succ y_B \mid x) = \sigma(r(x, y_A) - r(x, y_B)) \tag{1}$$

where $r(x, y) \in \mathbb{R}$ is the reward score and $\sigma$ is the sigmoid function. The reward model is trained to maximize the log-likelihood:

$$\mathcal{L}_{\text{BT}} = -\mathbb{E}_{(x,y_A,y_B,\delta)\sim\mathcal{D}}[\delta \log P(y_A \succ y_B \mid x)] \tag{2}$$

where $\delta \in \{-2, -1, 0, +1, +2\}$ is the annotator preference strength.

### 3.2 Multi-Head Architecture

Our reward model uses a Zen MoDE encoder to produce a contextual representation of $(x, y)$, with separate output heads for each alignment dimension:

$$r_d(x, y) = \mathbf{w}_d^\top h(x, y) + b_d, \quad d \in \{\text{help}, \text{harm}, \text{hon}, \text{inst}\} \tag{3}$$

where $h(x, y) \in \mathbb{R}^{d_{\text{model}}}$ is the [EOS] token representation from the final transformer layer. The combined reward is:

$$r(x, y) = \sum_d \lambda_d \cdot r_d(x, y) \tag{4}$$

with per-domain weights $\lambda_d$ learned on the validation set.

### 3.3 Ensemble Construction

We train 5 reward models with different random seeds and data orderings, then ensemble:

$$r_{\text{ens}}(x, y) = \frac{1}{5} \sum_{k=1}^{5} r^{(k)}(x, y) \tag{5}$$

Ensemble disagreement $\text{Var}_k[r^{(k)}(x, y)]$ serves as an uncertainty estimate: high-variance examples are flagged for additional human review during RLHF training.

## 4 Synthetic Preference Generation

### 4.1 Red-Team / Blue-Team Protocol

Human preference data is expensive and limited in coverage. We augment it using a red-team model trained to generate challenging prompts that expose alignment failures, and a blue-team (the base policy) that responds. The red-team generates prompts by:

1. Sampling a seed prompt $x_0$ from the training distribution.
2. Applying a perturbation $\Delta$ that targets known weak spots (jailbreaks, out-of-distribution topics, adversarial formatting).
3. Generating the perturbed prompt $x' = T(x_0, \Delta)$.

For each perturbed prompt, the base policy generates two responses with different decoding temperatures ($T_1 = 0.7$, $T_2 = 1.2$), producing a natural diversity in response quality. The reward model ensemble scores both responses and the pair is added to the preference dataset with the model scores as soft labels.

### 4.2 Quality Filtering

Synthetic pairs are filtered by:

- Ensemble agreement: $\text{Var}_k[r^{(k)}] < \sigma_{\text{thresh}}$ ensures the preference label is well-defined.
- Margin threshold: $|r_{\text{ens}}(y_A) - r_{\text{ens}}(y_B)| > m_{\min}$ ensures the pair is informative (non-trivial).
- Semantic diversity: the prompt $x'$ must differ from training prompts by at least 0.15 in embedding distance.

This process generates 8M synthetic preference pairs, of which 3.4M pass all filters.

## 5 Direct Preference Optimization (DPO)

DPO [5] directly optimizes the policy using preference data, bypassing explicit reward model training:

$$\mathcal{L}_{\text{DPO}} = -\mathbb{E}_{(x, y_w, y_l)} \left[ \log \sigma \left( \beta \log \frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta \log \frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)} \right) \right] \tag{6}$$

where $y_w$ and $y_l$ are preferred and dispreferred responses, $\pi_{\text{ref}}$ is the reference (SFT) policy, and $\beta$ controls the KL penalty.

We compare DPO trained on our human + synthetic preference data to full RLHF using PPO with our ensemble reward model.

# 6 Experiments

## 6.1 Reward Model Accuracy

Table 3: Reward model accuracy on held-out human preference pairs (binary: correct if reward model prefers the human-preferred response).

| Model | Accuracy (%) | Human corr. (Pearson $r$) |
|---|---|---|
| Single RM (no ensemble) | 77.8 | 0.778 |
| Ensemble (5 RMs) | 81.4 | 0.814 |
| Multi-head + ensemble | 85.2 | 0.852 |
| + Synthetic data | **87.3** | **0.873** |

## 6.2 RLHF vs. DPO Comparison

Table 4: Alignment benchmark results. AlpacaEval 2.0 win rate vs. GPT-4 Turbo. MT-Bench score (1–10). Arena Elo (Chatbot Arena).

| Training | AlpacaEval 2.0 (%) | MT-Bench | Arena Elo |
|---|---|---|---|
| SFT only | 18.4 | 7.8 | 1124 |
| DPO (human data) | 31.7 | 8.4 | 1198 |
| DPO (+ synthetic) | 36.2 | 8.7 | 1231 |
| RLHF (PPO, single RM) | 38.4 | 8.9 | 1248 |
| RLHF (PPO, ensemble RM) | **44.8** | **9.2** | **1312** |

Full RLHF with ensemble reward outperforms DPO by 8.6 pp on AlpacaEval. However, DPO at significantly lower compute cost achieves competitive performance on MT-Bench.

## 6.3 Reward Model Calibration

Table 5: ECE of reward model scores. Lower is better. The ensemble reduces miscalibration by 41% vs. single RM.

| Model | ECE | $\Delta$ vs. single |
|---|---|---|
| Single RM | 0.124 | — |
| Ensemble (5 RMs) | 0.082 | −33.9% |
| Multi-head + ensemble | 0.073 | −41.1% |

## 6.4 Downstream RLHF Results

Table 6: Downstream benchmark performance after RLHF. RLHF with ensemble RM provides consistent improvements without degrading base capabilities.

| Model | MMLU | GSM8K | HumanEval | TruthfulQA |
|---|---|---|---|---|
| Zen MoDE-72B (base) | 84.7 | 90.8 | 79.3 | 74.8 |
| + SFT | 84.9 | 91.2 | 80.1 | 76.3 |
| + DPO | 85.1 | 91.8 | 80.7 | 78.9 |
| + RLHF (ensemble RM) | **85.4** | **92.1** | **81.3** | **82.4** |

### 6.5 Reward Hacking Analysis

Table 7: Reward hacking incidence: fraction of RLHF checkpoints where reward model score increases but human preference decreases.

| Configuration | Hacking rate (%) | Detected by ensemble (%) |
|---|---|---|
| Single RM, no detection | 14.2 | — |
| Ensemble RM | 7.8 | 89.3% of single-RM hacking events |
| Ensemble + variance gate | 3.1 | 97.2% of single-RM hacking events |

The variance gate (flagging high-disagreement ensemble predictions) detects 97% of reward hacking events at the cost of rejecting 8% of legitimate training updates.

## 7 Discussion

### 7.1 Ensemble vs. Single Reward Model

The ensemble provides two benefits beyond accuracy: (1) calibration improvement via variance-based uncertainty estimation, and (2) reward hacking detection through disagreement signals. The computational overhead of running 5 reward models during RLHF rollouts is 5× for inference, but reward model inference is typically <10% of total PPO compute, making this tractable.

### 7.2 Synthetic Data Quality

The synthetic preference pairs improve accuracy by 2.1 pp (Table 3). Quality filtering is critical: unfiltered synthetic data degrades accuracy by 1.4 pp due to noisy margin pairs. The margin threshold $m_{\min}$ is the most important filter.

### 7.3 DPO Compute-Quality Trade-off

DPO avoids reward model training and PPO rollout generation, reducing alignment compute by approximately 8×. For deployment scenarios where compute is the primary constraint, DPO with synthetic preference augmentation is the recommended choice, achieving 81% of full RLHF performance at 12.5% of the compute.

## 8 Conclusion

We have presented the Zen alignment stack: multi-head ensemble reward models trained on 1.2M human preferences plus 3.4M filtered synthetic pairs, combined with RLHF or DPO fine-tuning. Key results: 87.3% reward model human correlation, 44.8% AlpacaEval win rate after RLHF, and 97% reward hacking detection with ensemble variance gating. The full stack is recommended for production deployments; DPO is recommended when compute is constrained.

## References


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